

Questions about the construction of our three education groups

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Status: Ready for Charles to forward. No message has been sent.

Scope: Clarification of the data already supplied; no new extract or analysis requested.

Message to the source

Thank you for supplying these education panels. We are preparing to examine whether students' school context and standing relate differently to entering postsecondary education and eventually earning a bachelor's degree. Before proceeding, we would like to confirm how the existing columns were constructed.

This request covers all three groups together:

- **National Education Longitudinal Study of 1988 (NELS:88):** `datasets/nels88/nels88_big_fish_panel.csv`, with outcomes labeled through 2000.
- **High School and Beyond (HS&B), 1980 sophomore cohort:** the Sophomore rows of `datasets/hsb80/hsb80_big_fish_panel.csv`, with outcomes labeled through 1986.
- **HS&B, 1980 senior cohort:** the Senior rows of that same file, also labeled through 1986.

The easiest answer may be the original extraction and recoding scripts, together with the relevant codebook entries. If those materials answer the questions below, there is no need to prepare a separate long narrative. Please identify differences between the three groups and mark anything uncertain rather than reconstructing it from memory.

1. Which source files and students are represented?

- What exact survey releases, files, and waves produced each group? What are the original student and school identifier variables?
- What inclusion/exclusion steps produced the supplied files, before their analytic-sample flags were applied? Please include treatment of missing tests, survey nonresponse, transfers, duplicate records, and any imputation or substituted values.
- For NELS, are these exclusively original eighth-grade participants, or are students added in later waves included? How were `study_entry_code` and `base_year_eligible` used?
- For HS&B, how were `SOQFLAG`, `SRQFLAG`, and `questionnaire_respondent` used? Were students without a particular follow-up response retained using information from another wave?

Why this matters: We need to state whose outcomes we are studying and distinguish original survey participation from our later analysis filters.

2. What exactly is the performance score in each group?

- Which original variable or formula produced `own_performance_raw`? Which subjects does it measure, and when was it measured? Does the definition differ between the sophomore and senior cohorts?
- How was `own_performance_z` standardized: which reference population, survey weights if any, and standard-deviation convention? Was standardization performed separately by cohort, and before or after exclusions?
- How were missing-value codes, noncompleted tests, and imputed scores handled?
- In NELS, what is the relationship between the main performance score and `base_reading_std_score` / `base_history_std_score`? A repository primer describes a reading/history composite, but their simple average does not reproduce the supplied main score. We are asking for the correct definition, not assuming the supplied score is wrong.

Why this matters: Our intended comparison is between students with similar prior measured performance. We need to know which performance and measurement date that means.

3. Which classmates make up each school environment?

- Which school wave defines `school_id` in each group? For NELS, please confirm the reported tenth-grade school assignment and how transfers or missing school identifiers were handled.
- Were peer measures constructed from all sampled students with valid performance in that school and cohort, or from a narrower group such as follow-up respondents? Are the HS&B cohorts always kept separate when calculating peers?
- Are peer averages unweighted or survey-weighted? Are the school sizes counts of sampled students rather than estimates of the full cohort?
- Please confirm the definitions of the leave-one-out mean (the average excluding the focal student), peer spread, relative-performance measures, percentile, and top-decile measures. Which population defines the global top-decile cutoff? How are ties handled?
- Our checks reproduce the supplied school counts, peer means, and midpoint treatment of percentile ties. Please provide the original recipe so we can preserve that construction when analyzing outcomes with different amounts of missing information.

Why this matters: We should not accidentally change a student's measured school environment by keeping only classmates whose later outcomes are observed.

4. What do the education outcomes mean?

For each group, please provide original source-variable names, category labels, question universes, and the mapping into the supplied binary outcomes. Include distinctions among a true "no," a legitimate skip, a missing response, and an imputed value.

NELS-specific questions:

- Is `degree_code_2000` the original `F4HHDG` variable? The supplied panel treats -3 as no postsecondary entry/no degree, and -9 as missing. Does -3 identify people without postsecondary experience in the source release used?
- Does `any_postsecondary_by_2000` capture attendance at any time through 2000, including attendance without earning a credential? Is it derived solely from the degree code or checked against an enrollment-history field?

- Which source produces `hs_completion_code_2000`, and does `high_school_diploma_by_2000` deliberately exclude general educational development credentials and certificates of attendance?

HS&B-specific questions, separately for sophomores and seniors:

- Which original variable produces `attainment_code_1986`? Is this highest credential earned, highest level attended, or another measure?
- The supplied `any_postsecondary_by_1986` is one for attainment codes 3 and above and zero for codes 1–2. Does that include students who attended college or vocational education but earned no postsecondary credential? In particular, could code 2 include someone whose highest credential is a high-school diploma but who also attended college?
- What qualifies as `two_year_or_higher_by_1986`? Please distinguish degrees, vocational certificates, and years attended. What qualifies as bachelor’s-or-higher?
- What is the precise observation date/window behind “by 1986”? Are both cohorts’ outcomes taken from equivalent questions in that wave, or combined across waves? We will preserve their different time since high school.

Why this matters: Attendance and credential completion are different events. We cannot use a highest-credential variable as an attendance indicator unless its categories actually support that distinction.

5. Which students and weights should support each type of analysis?

- Please provide the intended definitions and uses of `outcome_observed`, `analytic_sample_min5`, `analytic_sample_min10`, and, for HS&B, `analytic_sample_min5_weighted`.
- Our checks indicate that minimum-five and minimum-ten refer to total sampled students including the focal student, not five or ten other classmates. Please confirm whether these thresholds were construction conveniences or chosen for a specific methodological reason.
- Which original weights became `followup_weight_2000`, `followup_weight_1986`, and `base_year_weight`? Which longitudinal target population does each weight represent, and which is appropriate for relating baseline performance to the reported follow-up outcome?
- In the existing minimum-five HS&B samples, 677 of 12,184 sophomores and 935 of 9,874 seniors have nonpositive follow-up weights. Their outcomes are nevertheless recorded. What does a nonpositive weight mean in these files, and was retaining these students intentional for descriptive peer calculations or unweighted analyses?
- Are survey strata, original sampling-cluster identifiers, or replicate weights available in the source files used? Please identify them or point to the documentation; no new extract is requested at this stage. We do not assume the recorded high school is the original survey sampling cluster.

Why this matters: A student can contribute to a descriptive sample without necessarily having a valid weight for estimating a national population rate. We need to keep these uses distinct.

6. Which background fields can be used as prior information?

- What original variables and measurement dates underlie socioeconomic status, sex, race, and school characteristics? Please include special missing codes that may remain numeric rather than blank.
- For HS&B, when were `high_school_grades_code`, `high_school_program`, `gpa10`, and `gpa12` measured, and for which cohort are they applicable? Are they student reports or transcript measures?
- Were any of these variables reconstructed using later information? Are there known comparability limitations across the three groups?

Why this matters: We want to avoid adjusting for something measured after the school experience we are trying to understand, or treating an inapplicable code as a real measurement.

7. What documentation already exists for possible later augmentation?

We are first exhausting the existing data. If already known or documented, please point us to available college destinations, institution type/selectivity, application/admission records, transcript class rank, enrollment/degree timing, and the later HS&B sophomore follow-up. Please distinguish fields actually inspected in your source release from possibilities mentioned in general survey documentation, and note public-use versus restricted-use status where known.

If application/admission indicators are available, please distinguish directly reported events from derived indicators that infer them from subsequent enrollment. We would only request added fields after identifying a specific unanswered question. There is no need to conduct a new availability search or obtain additional data to answer this message.

What would be most useful to send back

- The construction scripts and source release/file names.
- Relevant original variable definitions, recodes, and survey-weight guidance.
- A brief note identifying cohort differences, known limitations, and anything not yet verified.

That should let us proceed with the existing panels while keeping later augmentation focused.

Preparation record

The questions above were prepared after inspecting both supplied CSV files and auditing NELS, HS&B Sophomore, and HS&B Senior separately. No outcome models were fitted and neither original file was modified. Matching internal arithmetic does not establish the original survey-variable semantics.

Internal evidence is retained under `education_analysis/`:

- NELS: `docs/source_audit/EDUCATION_20260928_NELS_source_audit_and_next_step.md` and `outputs/source_audit_20260928/`.
- HS&B program: `code/EDUCATION_20260928_hsb_source_audit.py`.
- HS&B checks, source/program fingerprints, execution metadata, and separate cohort outcome mappings: `outputs/hsb_source_audit_20260928/`.

HS&B checks reproduce the existing analytic samples (12,184 sophomores in 930 schools; 9,874 seniors in 885 schools), school sizes, peer counts, outcome-observed and analytic flags, and within-cohort percentile calculations. Peer means agree to within 0.00000550 raw-score

units. Bachelor's completion is nested within the supplied entry indicator, but that arithmetic alone does not establish whether the entry indicator includes attendance without a credential. The positive-weight analytic flags reproduce and retain 11,507 sophomores and 8,939 seniors. No weight exclusion or outcome recode has been adopted for new analysis.